# Paleomagnetic Investigations of the Tertiary and Quaternary Rocks: IX The Pliocene of Southeast Sicily and some Cretaceous Rocks from Capo Passero

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With 6 figures and 2 tables

**Zusammenfassung**

Die Aufsammlung und Untersuchung plio-pleistozäner und oberkretazischer Basalte in SE-Sizilien ergab die folgende paläomagnetische Information:

| Alter | D | J | $\alpha_{95}$ | Pol-Position | | dp | dm |
|---|---|---|---|---|---|---|---|
| Plio-Pleistozän | 161.0 | — 50.0 | 8.3 | 89 E<br>91 W | 73 S<br>73 N | 8.8 | 13.1 |
| Ober-Kreide | 162.9 | — 30.1 | 9.8 | 56 E<br>124 W | 64 S<br>64 N | 5.0 | 9.2 |

Unter der Annahme einer inversen kretazischen Feld-Richtung überlappt der Konfidenzkreis des virtuellen Pols den kretazischen magnetischen Pol für Afrika, der von MCELHINNY et al. an den Gesteinen des Mlanje Syenites bestimmt wurde.

Wenn SE-Sizilien zur Zeit der Extrusion dieser oberkretazischen Laven an Afrika „angeheftet" war, dann besagt dieses Ergebnis, daß eine extensive relative Bewegung zwischen diesen beiden schwerlich nacheinander stattgefunden hat.

**Abstract**

A sampling of Plio-Pleistocene and Upper Cretaceous basalts in SE Sicily yielded the following information:

| Age | D | J | $\alpha_{95}$ | Pole position | | dp | dm |
|---|---|---|---|---|---|---|---|
| Plio-Pleistocene | 161.0 | — 50.0 | 8.3 | 89 E<br>91 W | 73 S<br>73 N | 8.8 | 13.1 |
| Upper Cretaceous | 162.9 | — 30.1 | 9.8 | 56 E<br>124 W | 64 S<br>64 N | 5.0 | 9.2 |

Assuming that the Cretaceous direction is reversed, the confidence oval for the virtual pole overlaps that for the Cretaceous pole of Africa determined from the Mlanje Syenite by MCELHINNY et al. If SE Sicily was attached to Africa when these Upper Cretaceous lavas erupted, this result implies that extensive relative movement between the two is not likely to have occurred subsequently.

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**Résumé**

La compilation et l'analyse de basaltes d'âge plio-pleistocène et crétacique supérieur dans la SE-Sicile fournit l'information paléomagnétique suivante:

| âge | D | J | $\alpha_{95}$ | position de pôle | | dp | dm |
|---|---|---|---|---|---|---|---|
| plio-pleistocène | 161.0 | — 50.0 | 8.3 | 89 E<br>91 W | 73 S<br>73 N | 8.8 | 13.1 |
| crétacé supérieur | 162.9 | — 30.1 | 9.8 | 56 E<br>124 W | 64 S<br>64 N | 5.0 | 9.2 |

Dans l'acceptation d'un champ inversé au Crétacique, le cercle de confidence du pôle virtuel recouvre le pôle magnétique crétacique pour l'Afrique, tel que McElhinny et al. l'ont déduit des roches de la Syénite de Mlanje. Si l'on rattache le SE de la Sicile à l'Afrique au moment de l'éruption de ces laves du Crétacique supérieur, ce résultat signifie qu'il est peu probable qu'un mouvement relatif extensif ait eu lieu entre les deux.

**Краткое содержание**

Палеомагнитные исследования плио-плейстоценовых и верхнемеловых базальтов юго-восточной части Сицилии дали следующие результаты:

| возраст | D | J | α 95 | положение полюса | | dp | dm |
|---|---|---|---|---|---|---|---|
| Плио-плейстоцен | 161,0 | —50,0 | 8,3 | 89 E<br>91 W | 73 S<br>73 N | 8,8 | 13,1 |
| верхний мел | 162,9 | —30,1 | 9,8 | 56 E<br>124 W | 64 S<br>64 N | 5,0 | 9,2 |

Если предположить инверсию направления магнитного поля в меловом периоде, то конфидентный круг мнимого полюса в меловом периоде перекрывал магнитный полюс Африки, определенный Mc Elhinny и др. на породах Mlanje Syenites. Если южно-восточная Сицилия во времяна экструзии этих верхнемеловых лав «прикреплялась» к Африке, то наши данные говорят о том, что обширные относительные движения этих объектов вряд-ли следовали друг за другом.

## Introduction

Tertiary and Pleistocene volcanic rocks crop out over an area of about 400 km² in the Iblean Hills west of Syracuse (Fig. 1). A traverse in the valley of the Concerie at Vizzini indicated the following succession:

120 m Basalt flows: commonly columnar with vesicular tops. Locally intercalations of calcarenite found near the base.

70 m app.: Clastic volcanic glass with occasional pillows of basalt. White marl with pelecypods, Middle Miocene in age (Travaglia, 1885).

The lavas are predominantly submarine in origin and have been uplifted with little evidence of tilting or deformation. Inclinations of up to 15° have been recorded but these may reflect original, rather than tectonically imposed, dips.

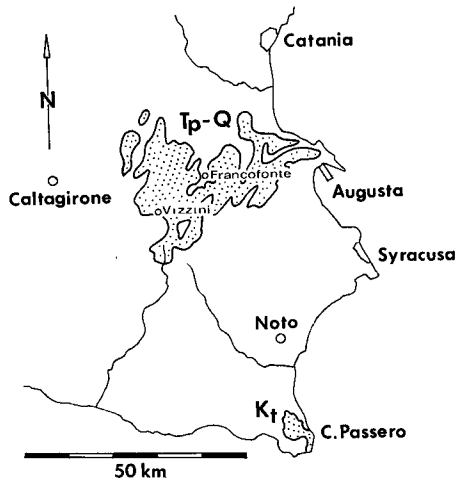



Fig. 1. Southeastern Sicily, showing outcrops of volcanic rocks. $T_p$—Q: Pliocene-Quaternary; $K_t$: Turonian.

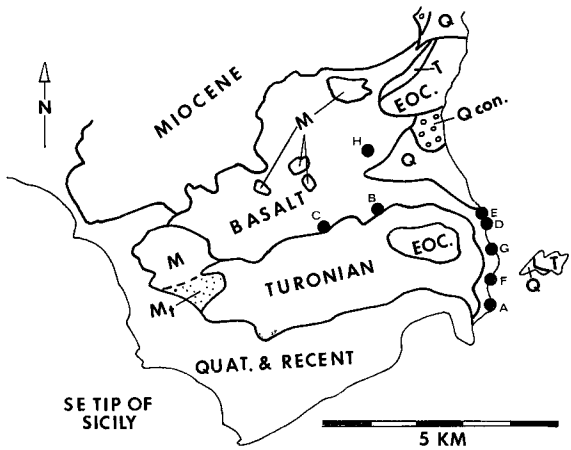



Fig. 2. Southeastern tip of Sicily. Geologic sketch map based on field observations and TRAVAGLIA (1886). Turonian, T: white limestone with Hippurites; Eoc.: white nummulitic limestone; Miocene, M: compact marly limestone, Langhian-Helvetian; $M_t$: sandy clay, Tortonian; Quat. & Recent, Q: sands and gravels; $Q_{con}$: gravelly and shelly calcareous deposits. A—H: basalt outcrops visited. A, B, E, G and H provided samples reported in this paper.

In places the thickness of lavas may exceed 200 m. To the east the basalt hills are truncated by marine terraces (CAMPIONE, 1961) upon which rest coquinas and calcarenite of Calabrian and Sicilian age. The flows are thus inferred to be of Pliocene age.

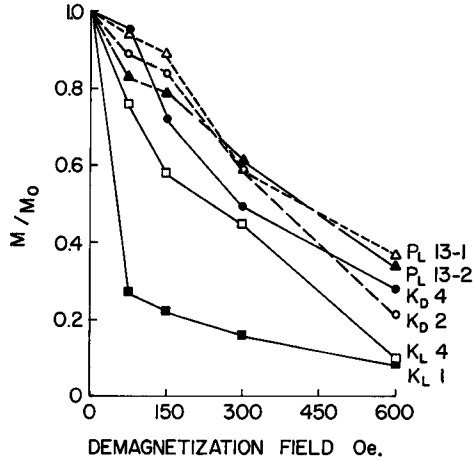



Fig. 3. Normalized demagnetization curves of samples of Pliocene lavas ($P_L$) and Cretaceous lavas ($K_L$) and dyke samples ($K_D$) from Sicily.

Further to the south at Capo Palo near Pachino on the extreme southeastern tip of Sicily the upper part of a series of submarine basalts and associated minor intrusions crop out (Fig. 2). The volcanic rocks are overlain by a white limestone rich in H i p p u r i t e s. The limestone dips at a low angle towards the east south east. In the lava series only two flows could be seen, however, numerous dykes occur and these were extensively sampled. The dykes are truncated by the limestone. The best estimate for the age of these volcanic rocks is Turonian, based upon the age of the limestone.

## Palaeomagnetic investigation

Approximately five block samples per flow or dyke were oriented in the field. From each of these a single core was drilled in the laboratory. After measurement of NRM (natural remanent magnetization using a FOSTER [1966] slow speed fluxgate spinner), a single core per igneous unit was progressively demagnetized in an alternating magnetic field. The maximum field strength applied, 600 oe, was reached in four stages (Fig. 3). Examination of plots of intensity against declination and inclination show that in the majority of cases the secondary component of magnetization is small, and removed in fields of either 150 oe or 300 oe. The remaining samples were then all demagnetized at these two field strengths. The direction adopted as the best representation of the field at the time of intrusion or extrusion was that at which the site mean had the lowest $\alpha_{95}$ and highest K values (see IRVING et al., 1961). These site mean directions of magnetization are shown in Fig. 4 a, b and the data is listed in table 1 a, b. In only a single case was it impossible to establish a stable end point to the demagnetization process, approximately one third of the sites however had circles of confidence in excess of 10° of which three were in excess of 20°.

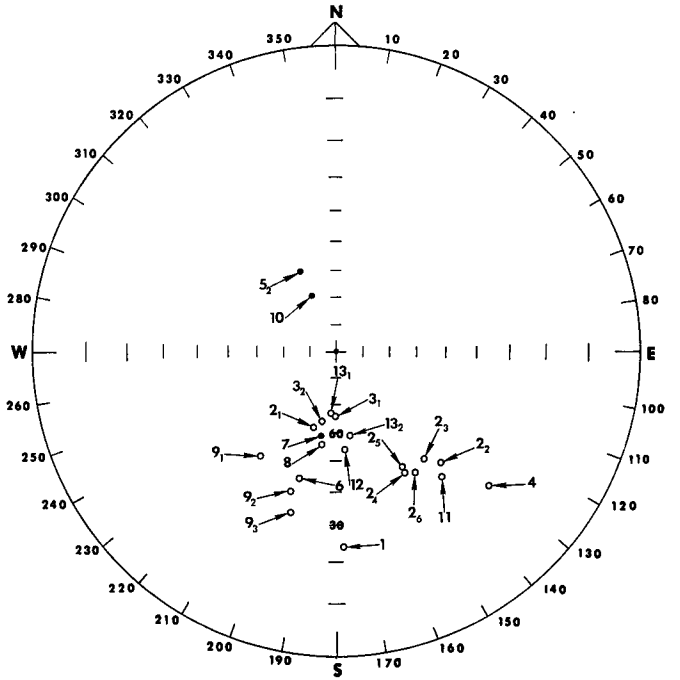
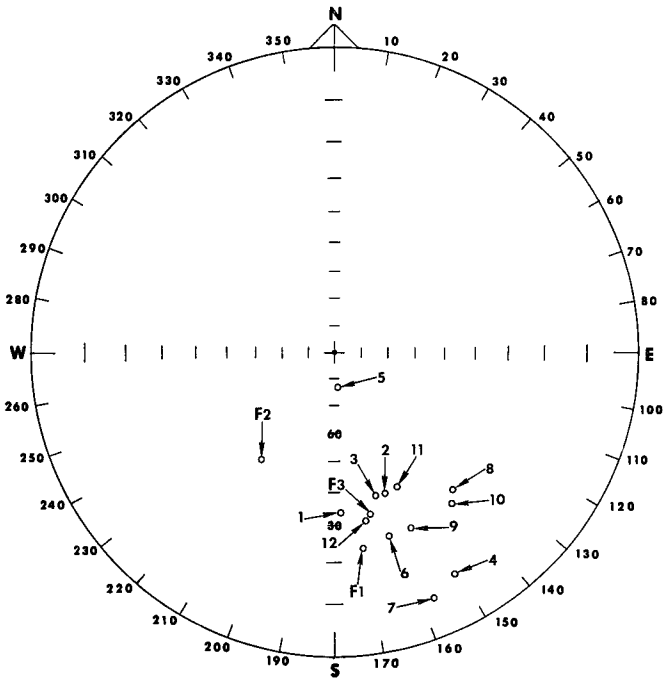



Fig. 4. Stereographic plots of the site mean declination and inclination values of: a) Pliocene lavas and b) Cretaceous lavas and dykes from south eastern Sicily.



## Ore-petrological investigation

The same cores used for the magnetic work were also used in the study of the ore petrology. Single cores from some twenty four sites were polished and studied in reflected light. On average the rocks contained some 1—8% of opaque ores by volume. Ilmenite, titano-magnetite, pyrite, chalcopyrite and chromite were identified together with minor amounts of geothite and lepidocrocite in some cases. The state of oxidation (BUDDINGTON & LINDSLEY, 1964) of the ore minerals was determined and following the techniques established by WILSON, HAGGERTY & WATKINS (1968), ADE-HALL, KHAN, DAGLEY & WILSON (1968) an oxidation number on the scale 1 to 6 was assigned based upon the titanomagnetites and 1 to 3 based upon the ilmenites. These numbers were based upon 300 grain counts. The results, and some notes for individual cores are listed by site in table 2 a, b.

In general in the more highly oxidized samples a general reddening of the sample could be observed. It is also of interest that as a generalization the Pliocene lavas are more highly oxidized than the Cretaceous rocks. Low temperature oxidation with the production of maghemite has also been observed, and this has proceeded to a greater extent in the latter.

## Discussion

A. Pliocene rocks: The majority of flows sampled were reversely magnetized (only two sites showed a normal magnetization). Since periods of normal and reverse polarity are of approximately the same duration during the Cenozoic, we take this to indicate that the flows sampled were extruded during a relatively short time interval. The virtual pole position computed from nineteen sites is given below:

$$D = 161.0 \quad I = -50.0 \quad \alpha_{95} = 8.3 \quad K = 22 \quad R = 17.591$$

virtual pole positions: 89 E 73 S (91 W 73 N) dp = 8.8, dm = 13.1.

B. Cretaceous rocks: The Cretaceous rocks were all reversely magnetized, a rather unusual event in view of the predominantly normal magnetization reported for the greater part of the period (HELSLEY & STEINER, 1969). The estimated age of the volcanics Turonian is sufficiently close to the lower Coniacian Kuldginskaja zone of Percheski (CREER, 1971) to suggest that if the magnetic zonation is valid the beds may be slightly younger than originally supposed and implies that the limestone too must be Coniacian or younger. Correlation with the nearest zone of reversed magnetization proposed by MCELHINNEY & BUREK (1971), the Santonian Akoh zone, is regarded as unlikely. The alternative is to suggest that the lavas are lower Cretaceous (Aptian/Albian) implying an important discontinuity between them and the overlying limestone.

The mean direction of magnetization listed below yields a virtual pole

$$D = 162.9 \quad I = -30.1 \quad \alpha_{95} = 9.8 \quad K = 13 \quad R = 14.377$$

virtual pole position: 56 E 64 S (124 W 64 N) dp = 5.0, dm = 9.2.

Table 1a. Summary of magnetic data
Sicily: Pliocene of Iblean region

| | | D | I | Sampling a/b | α 95 | K | R | Demag. field |
|---|---|---|---|---|---|---|---|---|
| Site 1. | | 178 | —24 | 5/5 | 56.8 | 3 | 3.600 | 600 oe.[1]) |
| Site 2. | 1st flow. | 198 | —58 | 5/5 | 5.3 | 300 | 3.990 | 150 oe. |
| | 2nd flow. | 137 | —36 | 4/4 | 5.8 | 250 | 3.990 | 77 oe. |
| | 3rd flow. | 140 | —40 | 4/4 | 15.6 | 36 | 3.916 | NRM |
| | 4th flow. | 150 | —40 | 4/5 | 9.8 | 89 | 3.966 | 150 oe. |
| | 5th flow. | 150 | —42 | 4/4 | 5.2 | 307 | 3.990 | 300 oe. |
| | 6th flow. | 147 | —38 | 5/5 | 4.8 | 259 | 4.985 | 150 oe. |
| Site 3. | 1st flow. | 182 | —67 | 5/5 | 3.9 | 379 | 4.990 | 150 oe. |
| | 2nd flow. | 188 | —63 | 5/5 | 6.4 | 146 | 4.973 | 300 oe. |
| Site 4. | | 131 | —22 | 4/4 | 7.7 | 145 | 3.98 | 150 oe. |
| Site 5. | 1st flow. | no result | | | dispersed results. | | | |
| | 2nd flow. | 333 | +58 | 3/4 | 30.1 | 18 | 2.808 | 150 oe. |
| Site 6. | | 196 | —42 | 2/2 | estimated | | — | 300 oe. |
| Site 7. | | 194 | +59 | 5/6 | 9.2 | 71 | 4.864 | 150 oe. |
| Site 8. | | 190 | —55 | 4/4 | 7.7 | 143 | 3.9789 | 300 oe. |
| Site 9. | 1st flow. | 216 | —43 | 4/4 | 13.3 | 49 | 3.938 | 150 oe. |
| | 2nd flow. | 198 | —37 | — | — | — | — | 150 oe. |
| | 3rd flow. | 196 | —31 | 2/2 | estimated | | — | 150 oe. |
| Site 10. | | 332 | +68 | 5/5 | 14.8 | 28 | 4.856 | NRM |
| Site 11. | | 140 | —33 | 5/5 | 15.2 | 26 | 4.848 | 150 oe. |
| Site 12. | | 174 | —53 | 5/5 | 9.2 | 41 | 4.943 | 150 oe. |
| Site 13. | 1st flow. | 185 | —68 | 5/5 | 6.5 | 138 | 4.971 | 300 oe. |
| | 2nd flow. | 172 | —58 | 5/5 | 8.7 | 79 | 4.949 | 300 oe. |

Table 1b. Summary of magnetic data
Sicily: Cretaceous of Pachino region

| | | D | I | Sampling a/b | α 95 | K | R | Demag. field |
|---|---|---|---|---|---|---|---|---|
| Dyke 1. | Site E | 178 | —34 | 5/5 | 5.1 | 225 | 4.982 | 300 oe. |
| 2. | Site E | 160 | —37 | 3/4 | 13.8 | 31 | 2.975 | 300 oe. |
| 3. | Site E | 164 | —37 | 4/4 | 12.0 | 60 | 3.950 | 300 oe. |
| 4. | Site E | 151 | —10 | 4/4 | 7.0 | 173 | 3.983 | 300 oe. |
| 5. | Site E | 177 | —77 | 4/4 | 4.1 | 491 | 3.994 | 150 oe. |
| 6. | Site E | 162 | —25 | 10/11 | 4.9 | 97 | 9.907 | 150 oe. |
| 7. | Site E | 157 | —7 | 5/5 | 14.0 | 31 | 4.869 | 150 oe. |
| 8. | Site G | 139 | —28 | 1/1 | — | — | — | 150 oe. |
| 9. | Site G | 156 | —25 | 8/8 | 4.0 | 195 | 7.960 | 150 oe. |
| 10. | Site G | 142 | —25 | 5/5 | 8.6 | 80 | 4.950 | 150 oe. |
| 11. | Site G | 152 | —38 | 6/6 | 6.1 | 120 | 5.958 | 300 oe. |
| 12. | Site B | 168 | —31 | 7/7 | 10.7 | 33 | 6.817 | 77 oe. |
| Flow 1 Capo Palo (F 1 Fig. 4b) | | 171 | —24 | 7/7 | 22.4 | 8 | 6.260 | 150 oe. |
| Flow 2 Capo Palo (F 2 Fig. 4b) | | 214 | —43 | 7/8 | 13.0 | 23 | 6.730 | 150 oe. |
| Flow 3 Cozzo S. Lucia (F 3 Fig. 4b) | | 167 | —32 | 9/9 | 15.3 | 12 | 8.351 | 150 oe. |

Table 2a. Summary of ore Petrology
Sicily: Pliocene of Iblean region

| | Oxidation M | No.[1] I | mean grain size $\mu$ | Notes |
|---|---|---|---|---|
| Site 1 | 4.8 | 1.9 | 42 | |
| Site 2—1 | ore pigment | | | unidentifiable |
| —2 | 5.06 | 2.24 | needles 287/14 | red colour around ores with high oxidation state |
| —3 | 4.5 | 1.8 | 191 | some rare maghematization |
| —5 | 5.06 | 2.12 | 100 | red colour around ores with high oxidation state |
| —6 | 5.3 | 2.4 | 128 | red colour around ores with high oxidation state ilmenite $>>$ titanomagnetite. |
| Site 3—1 | 1.0 | 1.1 | needles 225/23 | some rare maghematization |
| —2 | 2.65 | 1.3 | needles 225–450/22.5 | some rare maghematization ilmenite $>>$ titanomagnetite |
| Site 4 | 2.77 | 1.0 | 21 | formation of maghemite |
| Site 5—1 | 1.0 | 1.0 | 77 | random vermicular formation of maghemite, titanomagnetite |
| —2 | 1.1 | 1.0 | 45 | intergrowths of ilmenite — titanomagnetites |
| Site 7 | 2.56 | 1.3 | 26 | ilmenite — titanomagnetite intergrowth, titanomagnetite $>>$ ilmenite |
| Site 10 | 4.57 | 1.94 | needles 232/23 | formation of maghemite, ilmenite $>$ titanomagnetite |
| Site 12 | 5.05 | — | 34 | |

[1]) M oxidation number based upon titanomagnetite.
I ilmenite.

Table 2b. Summary of ore petrology
Sicily: Cretaceous of the Pachino region

| | Oxidation M | No. I | mean grain size $\mu$ | Notes |
|---|---|---|---|---|
| Dyke 7 | 1.0 | 1.0 | 37 | maghemite replaces 50% or more of titanomagnetite |
| 9 | 1.0 | 1.0 | 34 | maghemite replaces 50% or more of titanomagnetite, skeletal titanomagnetite, titanomagnetite $>>$ ilmenite |
| 10 | 1.0 | 1.0 | 53 | formation of vermicular maghemite |
| 11 | 1.0 | (1.0) | 7.5 | replacement of titanomagnetite nearly completely by maghemite |
| 12 | 1.0 | 1.0 | needles, and fine ore pigment | formation of maghemite |
| flow 1 | 1.0 | 1.0 | 25 | formation of maghemite, ore pigment |
| 3 | 1.0 | 1.0 | 170 | intergrowths of ilmenite — titanomagnetite |

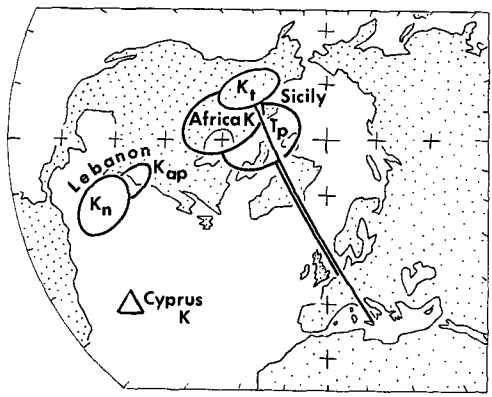



Fig. 5. Pliocene ($T_p$) and Upper Cretaceous ($K_t$) virtual poles for SE Sicily compared with the African Cretaceous pole from the Mlanje Syenite (Briden, 1967), Lebanese Neocomian and Aptian poles, and the Cretaceous pole for Cyprus. For discussion see text.

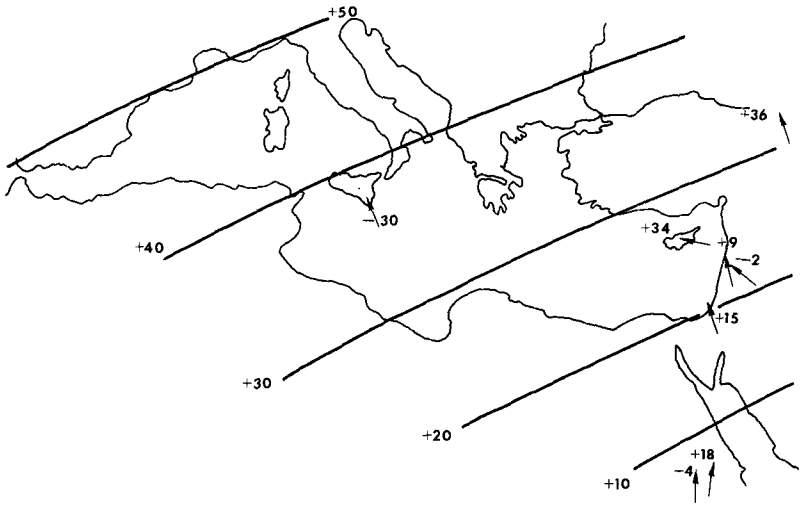



Fig. 6. A paleoisocline map of the Mediterranean area. The isoclines are based upon the mean virtual pole for the Cretaceous derived by Briden (1967) from African data. The declination measured for Sicily is shown by an arrow with the mean inclination value. For comparison other results from the Cretaceous rocks of the Mediterranean are shown. In each case the tip of the declination arrow approximates to the area of sampling.

position, the confidence limit of which intersects those computed for the Mlanje massif (Fig. 5) so the two cannot be regarded as significantly different. We may therefore conclude on palaeomagnetic grounds that during the Cretaceous, SE Sicily formed part of the African plate. This conclusion is a further example of

the consistency of plate tectonic configurations (e. g. Le Pichon, 1968; Morgan, 1968) and palaeomagnetic observations. Indeed, if we accept Sicily as part of the African plate a difference between the virtual pole positions is to be expected in view of the age difference between the two formations. The most striking illustration of the consistency of the African plate palaeomagnetic results is the indication of the concordance of the declination and inclination results on a palaeoisocline map (Fig. 6) based upon the Mlanje massif result. In addition to the result from Sicily there are measurements of Nubian Sandstone samples from two sites near Aswan (El Shazly & Krs, 1973). Both directions are figured, however, it is noticeable that the average of the two approximates very closely the predicted value.

Equally noticeable is the disagreement of the Cretaceous data from Cyprus (Vine & Moores, 1969) and the Lebanon (Gregor, Mertman, Nairn & Negendank, 1974). This suggests that in the present eastern Mediterranean, adjustment to the relative movement of the African and Eurasian plate has resulted in rotation or the generation of small adjustment plates.

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# Sedimentology and Genesis of the Cenozoic Sediments of Northwestern Himalayas (India)

By R. S. Chaudhri, Chandigarh *)

With 4 figures

**Zusammenfassung**

Herkunft und Sedimentationsraum der im Nordwestabschnitt des Himalaya anstehenden känozoischen Sedimente werden ausführlich dargestellt. Die Ergebnisse beruhen auf detaillierten sedimentologischen, einschließlich mineralogischen, petrographischen und petrochemischen Untersuchungen von mehr als 3000 repräsentativen Proben.

Es wird gefolgert, daß der Detritus der känozoischen Schichtglieder hauptsächlich von metamorphen, in den angrenzenden Himalaya-Gebieten anstehenden Gesteinen stammt.

Der Detritus der ältesten Einheit der känozoischen Folge wurde in marinen Flachwassern abgelagert. Die übrigen Schichten stellen Anhäufungen räumlich und zeitlich schwankender, nicht-mariner Ablagerungsräume dar.

**Abstract**

The paper discusses at length the provenance and the environments of sedimentation of the Cenozoic sediments exposed in the northwestern sector of the Himalayas. The results are based on detailed sedimentological (including mineralogical, petrographical and petrochemical) investigations of more than 3,000 representative samples.

It is concluded that the detritus of the Cenozoic formations was derived mainly from metamorphosed rocks exposed in the adjacent Himalayan regions. A comparatively smaller proportion of the sediments was contributed by acid plutonic, volcanic and sedimentary rocks.

The detritus of the oldest unit of the Cenozoic sequence was deposited in shallow marine waters. The remaining formations represent the accumulations of nonmarine environments fluctuating in space and time.

**Résumé**

La provenance des sédiments cénozoïques et le domaine de sédimentation du secteur nord-ouest de l'Himalaya sont décrits en détail. Les résultats reposent sur des recherches sédimentologiques, à la fois minéralogiques, pétrographiques et pétrochimiques, de plus de 3000 échantillons représentatifs. Les résultats montrent que les détritiques pro-

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